Question Bank – Computational Theory – PT 2

Q. 1. Design PDA to accept language $\{ a^{n-1} b^{2n+1} \mid n >= 1 \}$

Q. 2. Design PDA that check well-formed parentheses

Q. 3. Construct PDA accepting following language

$L = \{ a^n b^m c^n \mid m, n >= 1 \}$

Q. 4. Design PDA that accepts odd palindromes over {a, b, c}, where c exists only at the center of every string

Q. 5. Construct PDA accepting the language

$L = \{ a^{2n} b^n \mid n >= 0 \}$

Q. 6. Design Turing Machine to accept strings of type $L = \{ 0^n 1^n \mid n >= 1 \}$

Q. 7 Design Turing Machine to accept the language given by a regular expression

$R = 0 (0 + 1)^* 11$

Q. 8 Design a TM which recognizes words of the form

$\{ a^n b^n c^n \mid n >= 1 \}$

Q. 9 Design a TM for converting a input binary number to its one's complement of a binary number

Q. 10 Explain Universal Turing Machine

Q. 11 Design Turing Machine to add two Unary Numbers. Show simulation of the machine

Q. 12 Convert the following grammar to CNF.

$S \rightarrow ASB \mid \varepsilon$

$A \rightarrow AaS \mid a$

$B \rightarrow SbS \mid A \mid bb$

Q. 13 Convert the following grammar to CNF.

$S \rightarrow bA \mid Ab$

$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow aBB \mid bS \mid b$

Q.14 Convert the following grammar to GNF.

$S \rightarrow AA \mid a$

$A \rightarrow SS \mid b$

Q. 15 Convert the following grammar to GNF.

$S \rightarrow ABAb \mid ab$

$B \rightarrow ABA \mid a$

$A \rightarrow a \mid b$

Q. 16 Convert the following grammar to CNF AND GNF.

$S \rightarrow BS \mid Aa$

$A \rightarrow bc$

$B \rightarrow Ac$

Q. 17 Explain halting problem.

Q. 18 Compare recursive and recursively enumerable languages

Q. 19 Write short note on: Rice's Theorem

Q. 20 Write short note on: Post Correspondence Problem